

Modeling and Characterization of Large-Scale Wi-Fi Traffic in Public Hot-Spots

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Outline

- Goals
- An Overview of Data
- Arrival Count Modeling
- Connection Duration Modeling
- Simultaneous Users Modeling
- Conclusions

Motivation

Free WiFi Coming to Starbucks, Now a Seat Will be Harder to Find

Source: [AP](#) [Ap](#) • June 14, 2010

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In an effort to increase sales and bring in more customers, Starbucks will now be offering free WiFi service as of July 1st.

Starbucks Corp. will begin offering unlimited free [wireless](#) Internet access at all company-operated U.S. locations starting July 1, part of an ongoing effort to bring more customers in the door.



On average, [laptop](#) users spend about an hour using the wireless Internet in Starbucks stores while mobile phone users who can use Wi-Fi spend about 15 minutes on the network.

After the two-hour window, consumers at the Seattle chain were charged \$3.99 for two additional hours.

Officials said Monday that access will continue to be offered through AT&T. But it won't require a Starbucks loyalty card, according to the announcement Monday by CEO Howard Schultz, who spoke at a [conference](#) in New York.

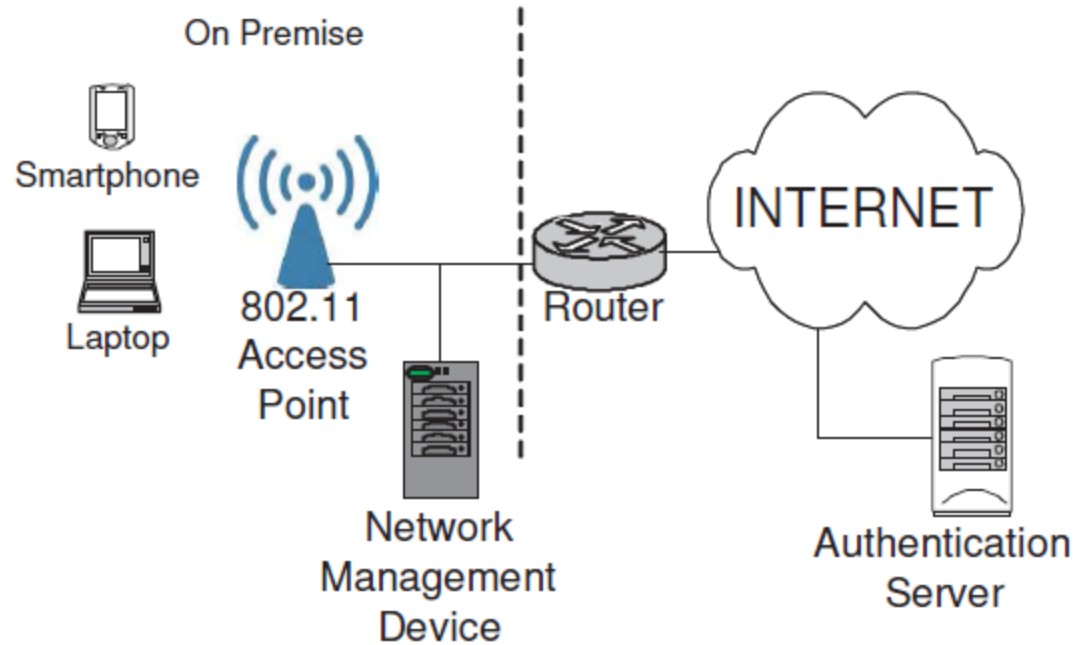
❑ Increasing number of **WLAN** deployments to meet the **growing demand** of (mobile) users for **wireless access**

Goals

- Study and analyze Wi-Fi traces collected by AT&T in March 2010 in NYC and SF
 - Coffee shops, fast food chains, book stores, hotels, stadiums
 - Data contains:
 - Connection login/logout times
 - Bytes uploaded/downloaded
 - Venue size (small, medium, large), zip codes, ...
- Realistic modeling of
 - Session arrivals
 - Connection duration distribution
 - Simultaneously present customer distribution



Data Collection

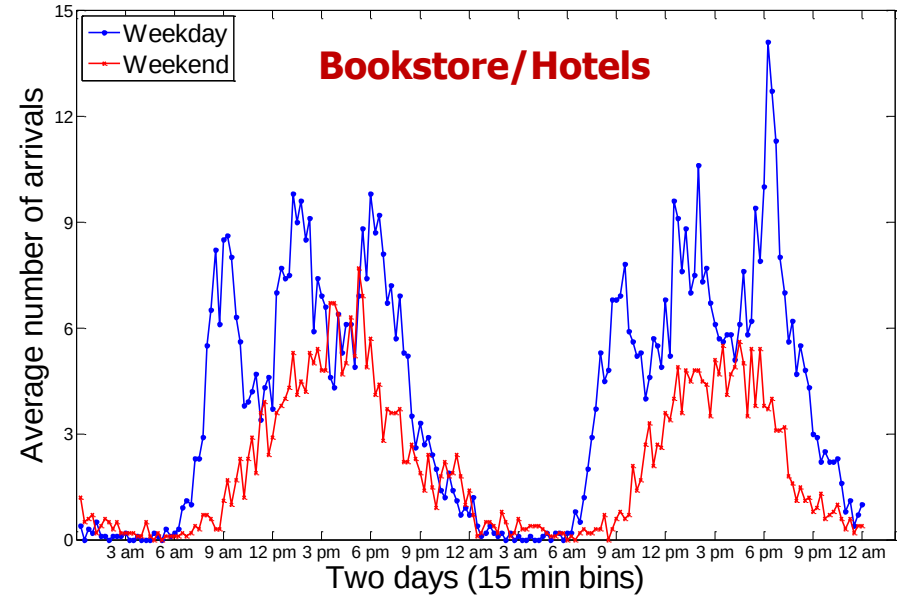
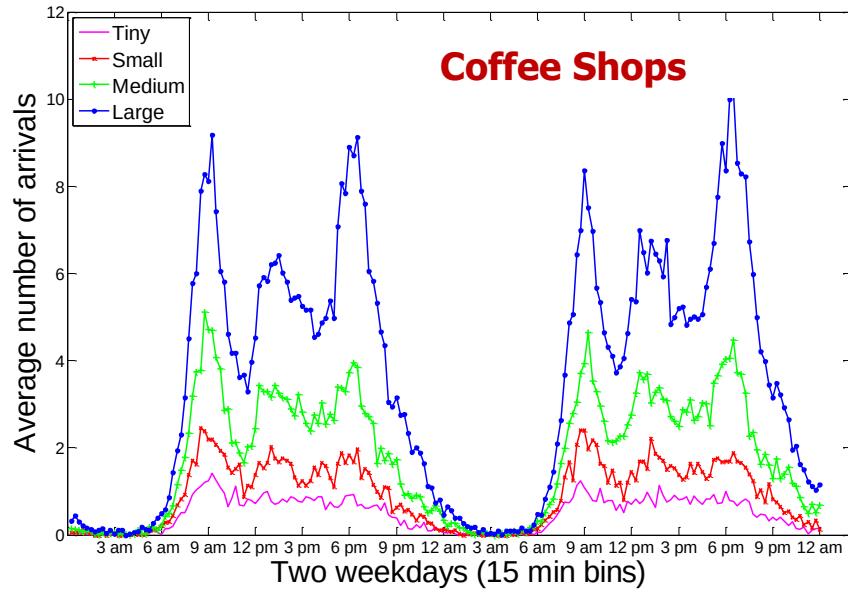


Mobile Internet Access using Wi-Fi Hotspots

Data Statistics

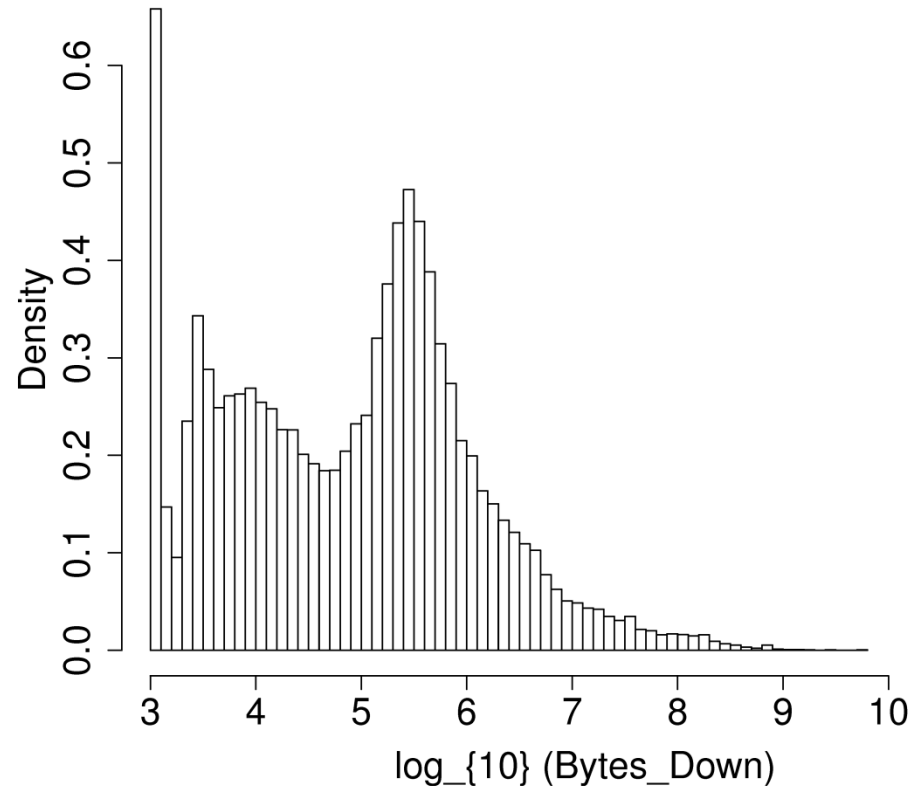
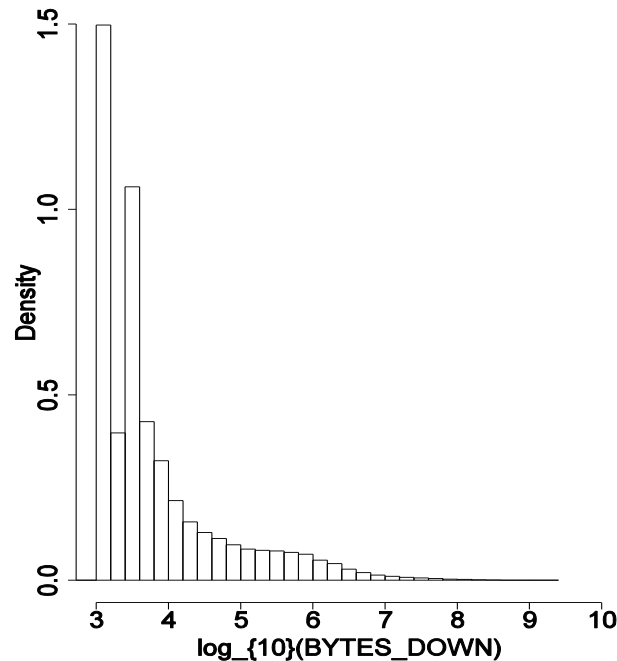
# of customers	234,742
# of devices	10
# of connections	1,322,541
# of cities	2 (NYC, SF)
# of Wi-Fi venues	362
# of zip codes	87
Trace duration	4 weeks (3 weeks training, 1 week validation)

Overview: Arrivals



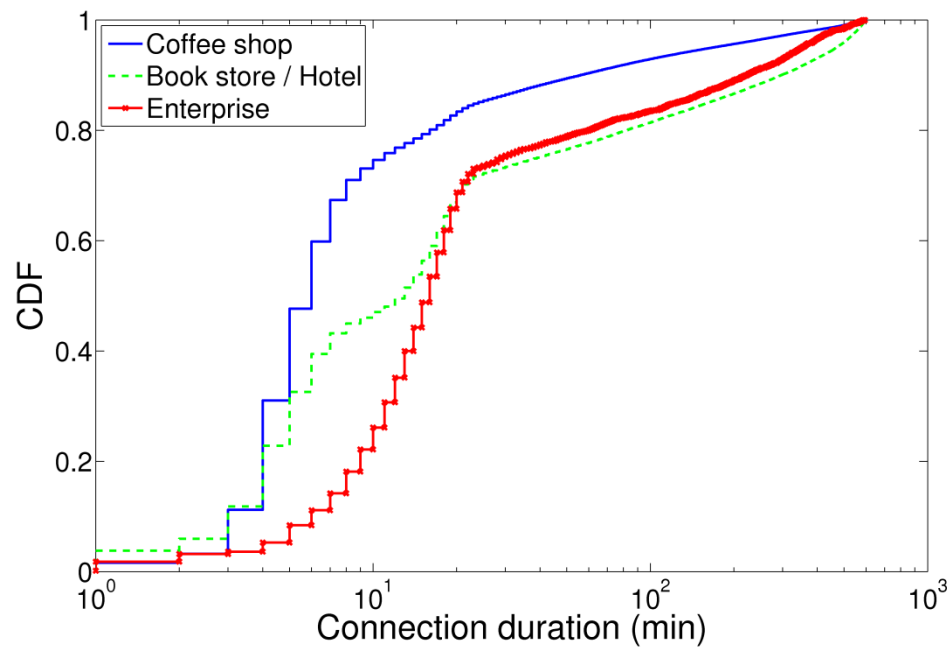
- Arrival rates **vary drastically** within the same business type
- **Characteristic peaks** in means across all categories within same business type
- Significantly different weekday and weekend patterns

Overview: Byte Counts

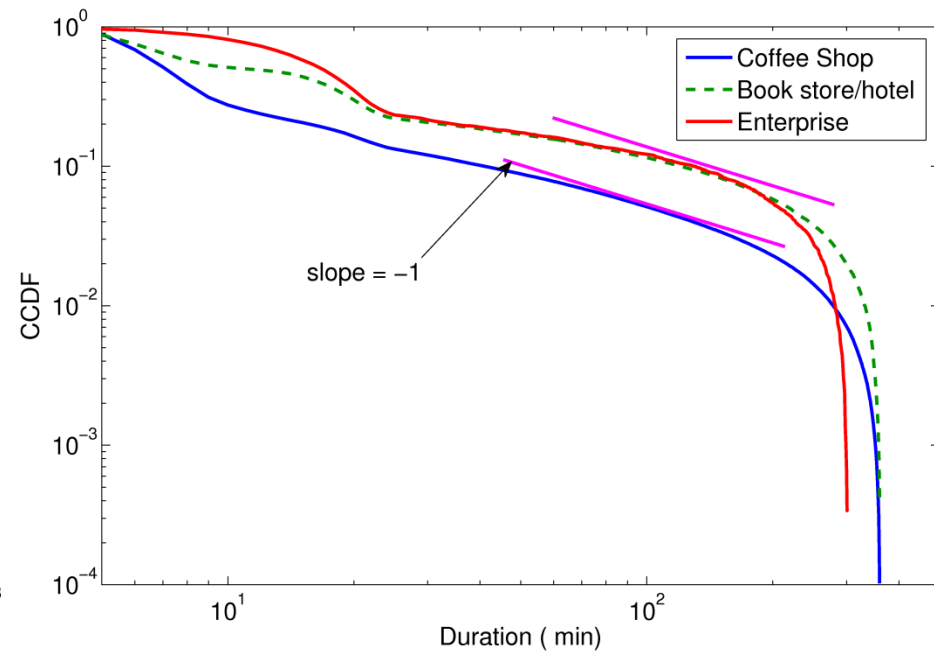


- Coffee shops: typically download a few KB
- Enterprises: typically download a few MB to a few GB
- Long tails

Overview: Durations



CDF of connection durations by business types



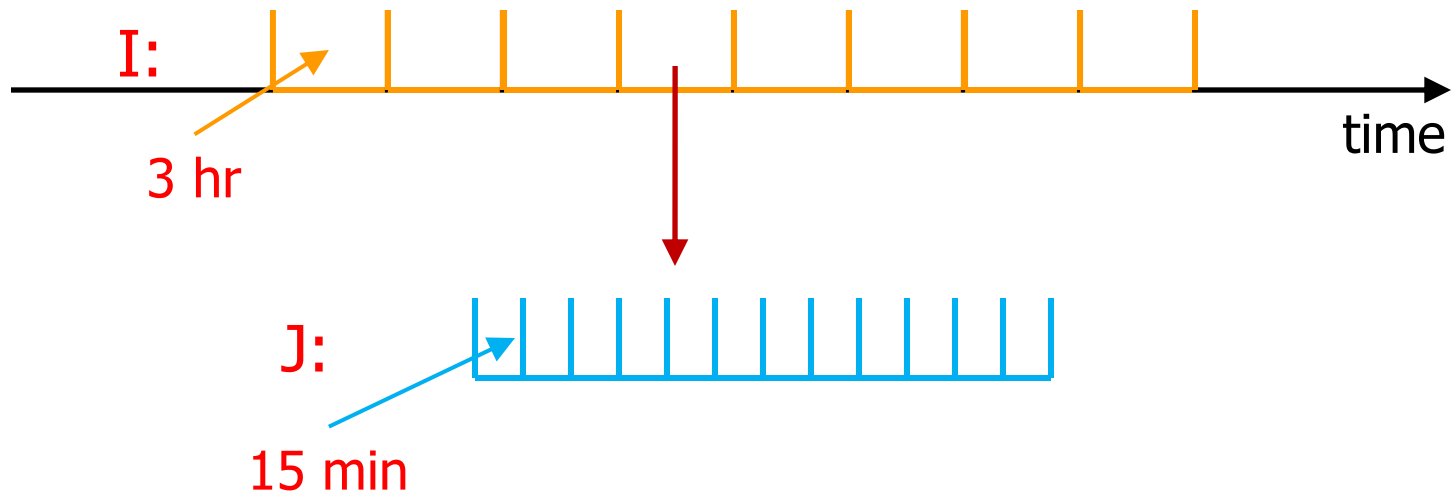
Complimentary distribution function of connection durations (log-log scale) by business types
=> Long tails

Arrival Count Modeling: Approach

- Data showed time-dependent arrival rates
 - MMPP fails
 - Models arrival counts with constant periods of arrival rate
 - Polynomial curve fitting to the observed mean
 - Poor performance
 - Could not capture within-day pattern with small no. of terms
 - Standard Poisson regression fails
- Non-homogeneous Poisson regression with clustering

Arrival Count Modeling

- Non-stationary Poisson Process
 - Time-dependent deterministic arrival rate
 - Divide time into 3 hour bins I: 8 bins per day
 - Divide each bin into 15 min slots J: 12 slots per bin



Arrival Count Modeling

- Poisson Regression Model (GLM)

- Polynomial type dependence on bin and slot numbers

$$\log(\lambda(I, J)) = \alpha + \sum_{k=1}^3 \beta_k I^k + \sum_{k=1}^3 \gamma_k J^k + \delta(I * J)$$

- First term

- Over-a-day mean behavior

- Sum terms

- Differential effects of specific cluster and slots within it

- Last term

- Interaction term – differential effect of slot J does not have to be the same across all clusters

Arrival Count Modeling

■ Clustering

□ K-means clustering:

- Cluster time slots into groups such that within each group the average number of arrivals do not differ much

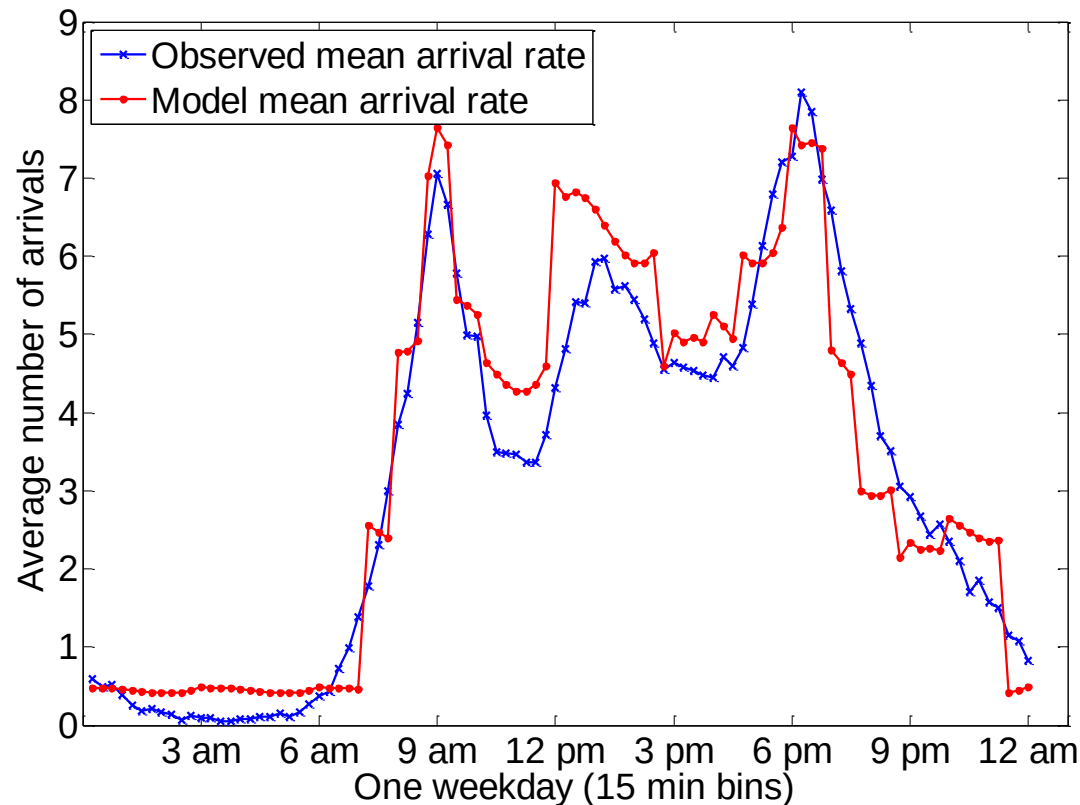
□ Automatic 24 hour wrap-around in clustering

Cluster	Slots
0	1–28, 94–96
1	83–87
2	32–34, 38–40, 64–66
3	35–37, 72–75
4	48–58
5	41–47, 59–63, 67–71, 76–78
6	79–82
7	29–31, 88–93

- Clusters of 15 min time slots over a day

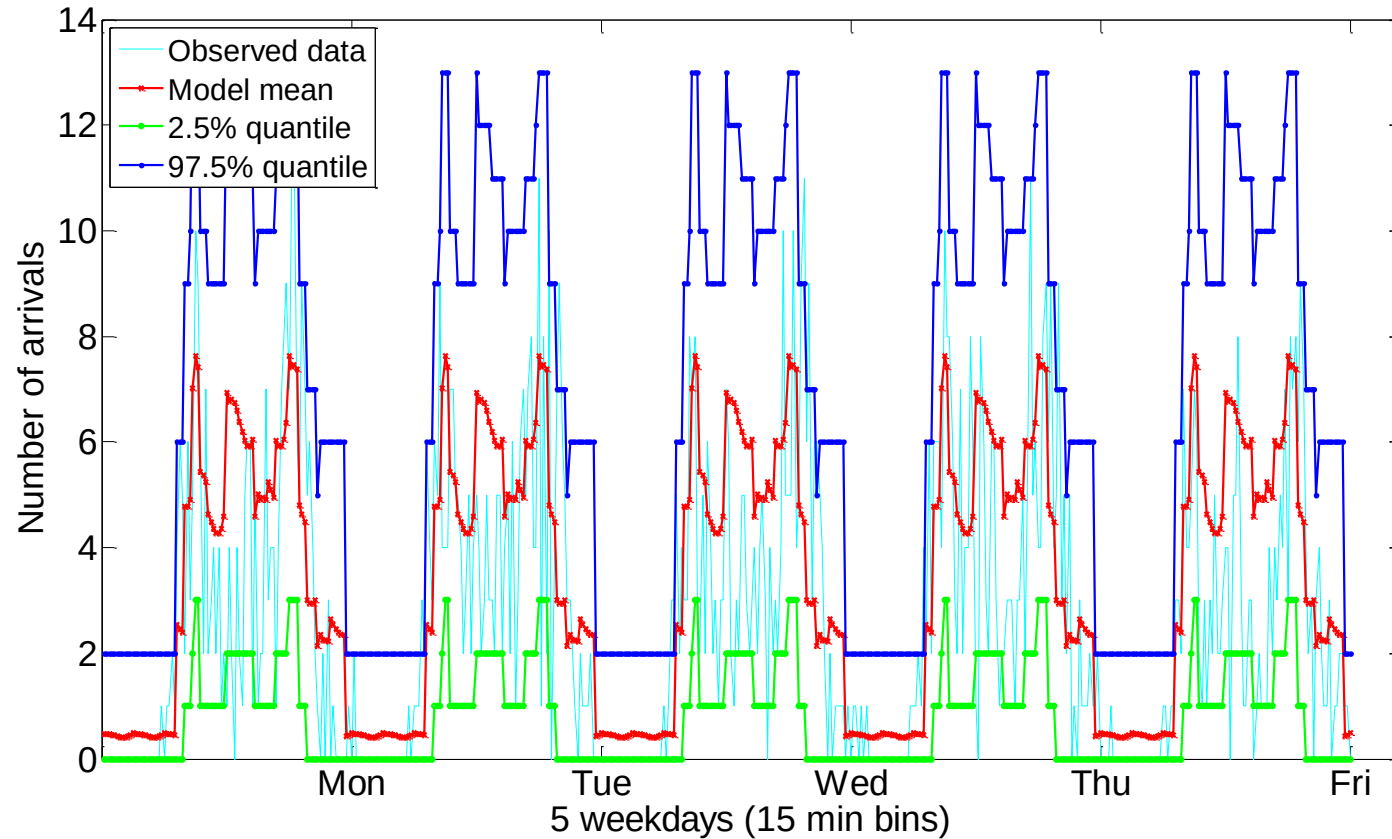
- Non-contiguous busy slots (35–37, 72–75) map to a common cluster

Results: Arrivals



☐ **Coffee shops:** Observed mean arrival rate plotted against the model mean arrival rate; these provide intra--day patterns for a cluster by averaging over its members

Results: Arrivals



□ **Coffee Shops:** Model mean arrival rate along with the 97.5% quantile and 2.5% quantile bands plotted against 5 days of validation data for an example coffee shop.

Session Duration Modeling

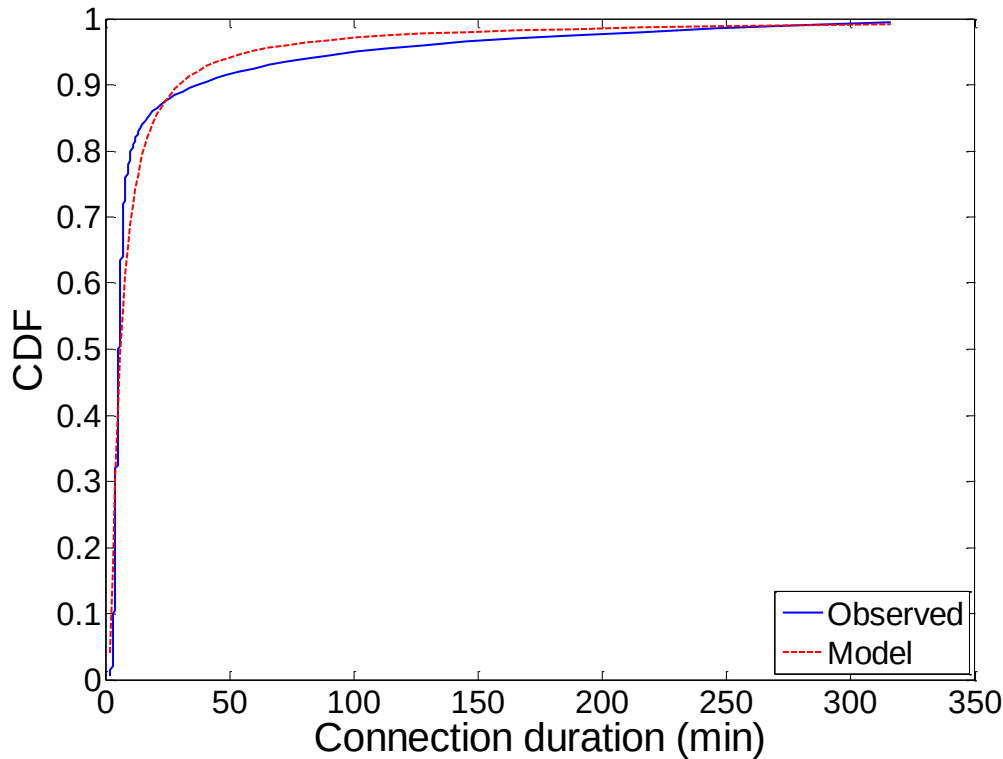
- Model the **logarithm of duration** (Y) as a **Phase-Type** (PH) random variable (X)

$$Y \sim e^X$$

PH-Type Distribution

- Properties of a PH-Type random variable
 - Distribution time to absorption in a Markov Process
 - Dense in the class of all distributions
 - Exponentially decaying tail asymptotically going to 0 as $e^{-\eta x}$, η is the real Eigen value of the rate transition matrix
 - Captures both tails and heads, as opposed to Pareto and Weibull

Results: Duration



□ Phase type distributions were fit using the EM algorithm

□ A fit of order 5 was found to be adequate

□ **Coffee Shops:** CDF plot of durations for coffee shops and data (truncated at 6 hours)

Simultaneous Connections

- Arrivals

- Non-homogeneous Poisson process (time-dependent arrival rates)

- Durations

- PH-type distribution

- Simultaneously present customers

- Queuing model

$$M_t/G/\infty$$

Simultaneous Connections

■ Theorem

The number of **busy servers** $Q(t)$, i.e., the number of simultaneously present customers, at time t follows a **Poisson distribution** with **mean** $m(t)$ given by:

$$m(t) = \int_0^t \lambda(\tau) [1 - H(t - \tau)] d\tau$$

where $H()$ is the service time distribution

Simultaneous Connections

- Novel proof based on semi-regenerative argument
 - Does not require the system to be empty at some infinite past
 - Simple, transparent, and general
- Show that the Probability Generating Function $G(t)$ of $Q(t)$ is

$$\begin{aligned} G_t(z, 0) &= E[z^{Q(0,t)}] \\ &= e^{(z-1) \int_0^t \lambda(\tau) \bar{H}(t-\tau) d\tau} \end{aligned}$$

Simultaneous Connections

■ Proof idea:



- No arrivals in $(u, t]$
- First arrival occurs at some v in $(u, t]$
- $Q(u)$: num of customers who arrive in $(u, t]$ and are still there at t
- $G(z, u)$: PGF of $Q(u, t)$
- Expected number of arrivals in $(0, t]$

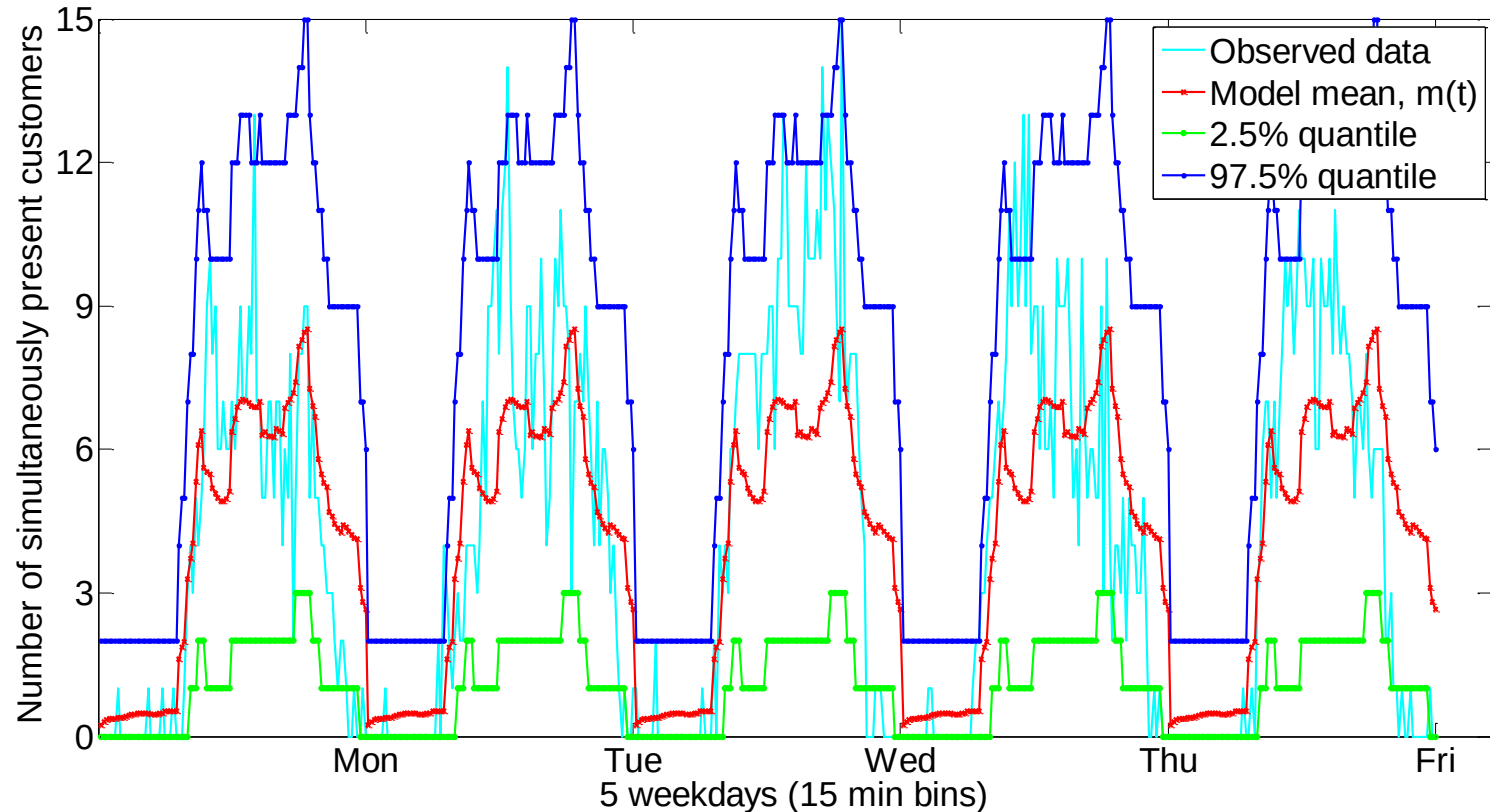
- Let
$$\Lambda(t) = \int_0^t \lambda(\tau) d\tau$$

Simultaneous Connections

- Solve the integral equation

$$\begin{aligned} G_t(z, u) &= E[z^{Q(u, t)}] \\ &= e^{-[\Lambda(t) - \Lambda(u)]} + \\ &\quad \int_u^t \lambda(v) e^{-[\Lambda(v) - \Lambda(u)]} \cdot \\ &\quad \{H(t - v) + z[1 - H(t - v)]\} G_t(z, v) dv \end{aligned}$$

Results: Simultaneous Connections



☐ **Coffee Shops:** Expected number of simultaneously present customers along with the **97.5% quantile** and **2.5% quantile** bands plotted against 5 days of validation data for an example coffee shop.

Conclusions

- Examined salient differences w.r.t.
 - Arrival counts, temporal variations, connection durations, byte counts
- Modeling
 - Arrival count modeling using statistical clustering and **non-stationary Poisson** model under GLM framework
 - Use of **Phase-Type** r.v. to model the logarithm of long-tailed durations
 - Simultaneously present customer modeling using a $M_t/G/\infty$ **queuing** model
- New proof on **semi-regenerative** argument for the number of busy servers in queue

- ❑ Amitabha Ghosh, Rittwik Jana, V. Ramaswami, Jim Rowland, and N. K. Shankaranarayanan, [Modeling and Characterization of Large-Scale Wi-Fi Traffic in Public Hot-Spots](#), *INFOCOM 2011*, Shanghai, China, April 2011.

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Thank you!